

Street walkability inference with street view imagery through fine-tuning and prompting pre-trained large models

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Shijia Lin^{1,2} , Xinyi Niu^{1,2} , Siwei Zhou^{1,2} , Yeyoumin Tian^{1,2} ,
Chenyi Wang^{1,2}  and Mingqi Shu^{1,2} 

Abstract

Large-scale assessment of street walkability using street view imagery is essential for supporting urban spatial governance and achieving sustainable development goals. This study proposes a walkability inference framework based on pre-trained multimodal large models, which integrates domain-specific knowledge of the hierarchy of walking needs through fine-tuning and prompt engineering. This framework consists of three sequential steps: filtration of street view images, hierarchical diagnosis of street walkability, and treatment for walkable streets. First, LoRA fine-tuning is applied to align the model's spatial perception standards with those of human experts, enabling independent assessment of accessibility, safety, comfort, and pleasurability, and facilitating the identification of spatial issues at each level. Second, prompt engineering is employed to embed the hierarchical logic of walking decision-making, allowing the model to determine priority improvement levels for the given street pedestrian environment and generate tailored spatial recommendations. To validate this approach, we conducted an empirical study, examining 17,937 street view images within the Inner Ring Road of Shanghai. The results confirm that our walkability inference model effectively performs logical inferencing based on the hierarchy of walking needs theory, addressing interpretability gaps and hierarchical deficiencies in street walkability assessment. Furthermore, the model generates context-aware, expert-level spatial recommendations in natural language, addressing the large-scale engineering requirements for improving pedestrian environments.

¹College of Architecture and Urban Planning, Tongji University, Shanghai, China

²Key Laboratory of Spatial Intelligent Planning Technology, Ministry of Natural Resources, Shanghai, China

Corresponding author:

Xinyi Niu, College of Architecture and Urban Planning, Tongji University, Rd. Siping 1239, Yangpu District, Shanghai 200092, China.

Email: niuxinyi@tongji.edu.cn

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Keywords

walkability, pre-trained large model, street view imagery, hierarchy of walking needs

Introduction

A walkable built environment enhances urban quality of life and promotes sustainable development by providing social and environmental benefits (Fonseca et al., 2021). Defined as the ease of walking in a given area, walkability reflects how individuals perceive and interact with the built environment and serves as a key indicator of the environment's potential to encourage walking (De Vos et al., 2023). The built environment influences pedestrians' willingness to walk at both the neighborhood and street scales. Neighborhood walkability, shaped by mesoscale urban form factors like population density, land use diversity, and street connectivity, influences the potential utility of walking trips (Wang and Yang, 2019). Street walkability, on the other hand, shapes pedestrians' subjective experiences through visual spatial features, like greenness, enclosure, and human scale (Park et al., 2015). Empirical studies suggest that street walkability is a stronger predictor of walking mode choice than neighborhood-scale attributes (Koo et al., 2022), making it a focal point for urban planners, spatial designers, and policymakers (Alidoust et al., 2018).

To better support spatial governance aimed at enhancing walkable environments, large-scale walkability assessments have become a mainstream research approach, enabling the identification of areas in need of improvement. Advances in street view imagery (SVI) and computer vision (CV), particularly deep learning (DL) technologies, have enabled scalable and precise assessments of street walkability (Yang et al., 2024). These tools allow the systematic detection of spatial disparities and spatiotemporal trends (Wang et al., 2024), facilitating evidence-based planning (He and He, 2023). However, existing comprehensive scoring methods often fall short in suggesting tailored improvement strategies at the street level, as they tend to overlook the distinct spatial barriers to walking that may underlie identical walkability scores. In other words, while these methods reveal the degree to which a street is walkable, they fail to explain why it underperforms and how its walkability can be improved.

The primary goal of walkability research is to identify deficiencies in the pedestrian environment and propose appropriate spatial interventions to address them (He and He, 2024). In this regard, Alfonzo's hierarchy of walking needs provides a robust theoretical framework for understanding how built environment features affect spatial perception and walking decisions (Alfonzo, 2005). This "pedestrian environment–spatial perception–walking decision" logic forms the cognitive basis for how human experts diagnose spatial issues and formulate design interventions. While recent AI methods enable large-scale assessments, they often fail to account for rich spatial context due to their reliance on composite indices. This highlights a key challenge: how can machines emulate human inference grounded in the hierarchy of walking needs to conduct qualitative spatial diagnosis and generate customized recommendations?

This study addresses the gap between walkability assessment and actionable improvement by fine-tuning and prompting pre-trained large models (PLMs) to perform expert-level inference. Leveraging their multimodal reasoning and natural language generation capabilities, the proposed approach integrates domain-specific knowledge into a walkability inference framework. This enables large-scale, context-aware diagnosis of pedestrian environments and the generation of tailored spatial improvement recommendations aligned with the hierarchy of walking needs.

Literature review

Pedestrian environment and walking behavior

A widely accepted consensus has emerged that the built environment significantly influences individuals' subjective perceptions and physical activity (Handy et al., 2002). Consequently, improving pedestrian environments to promote walking behavior and active modes of transportation has become a central concern (Gan et al., 2021). Under the framework of socioecological models, Alfonzo (2005) introduced the hierarchy of walking needs, providing a theoretical foundation for understanding how the built environment shapes walking decisions.

Inspired by Maslow's hierarchy of needs, walking needs can be classified into five levels: feasibility, accessibility, safety, comfort, and pleasurability. Among these five levels, while feasibility is primarily limited by an individual's physical ability, the remaining four levels are all shaped by environmental factors. Moreover, the theory explicitly posits a hierarchical relationship among these perception levels: only when basic needs are met or partially met will individuals consider higher-level needs. This implies that if basic needs are not fulfilled, even the satisfaction of higher-level needs cannot enhance the willingness to walk or promote walking behavior. This theory posits that individuals' walking behaviors are hierarchically influenced by different walking needs, with each perception level differentially shaped by the built environment.

Based on the inference logic of "pedestrian environment–spatial perception–walking decision," two methodological paradigms have emerged for measuring walkability. The subjective approach relies on self-reported surveys to assess pedestrians' willingness to walk and their multilevel spatial perceptions (Root et al., 2017). The objective approach uses measurable built environment indicators as proxies for walkability, such as greenery, enclosure, and diversity (Koo et al., 2022). While subjective methods yield perceptual insights, objective methods reduce biases inherent in descriptive approaches and allow for more comprehensive assessments (Yin and Wang, 2016). Nevertheless, early microscale studies were still constrained by labor-intensive manual analyses (Ewing and Handy, 2009).

Street view imagery and street walkability assessments

The advent of online SVI, coupled with advancements in CV, has revolutionized walkability research. SVI overcomes the limitations of field surveys or self-collected street imagery by providing cost-effective, high-resolution street-level data with human-eye perspectives (Li et al., 2022a). Furthermore, CV technologies automate feature extraction from digital images (Trucco et al., 1998), eliminating reliance on manual auditing (Yang et al., 2024). To be more specific, semantic segmentation enables systematic extraction of various visual factors (e.g., trees, buildings, and sky) from SVI, realizing the comprehensive quantification of hierarchical walking needs (He and He, 2024; Song et al., 2025). Notably, SVI-derived pedestrian-oriented metrics (e.g., greenery and connectivity) demonstrate superior explanatory power for walking behavior compared to satellite or road network data (Hajrasouliha and Yin, 2015; Ye et al., 2019).

Beyond objective metrics, SVI and CV facilitate large-scale prediction of subjective perceptions (Naik et al., 2014). By coupling annotated SVI with sentiment analysis, researchers decode nuanced perceptual experiences at scale and identify environmental determinants across perception levels (Zhang et al., 2018). This approach further reveals different walking preferences across different regions (Arellana et al., 2020) and specific populations (Cohen et al., 2024), offering localized support for walkability improvement tailored to pedestrians' actual needs. Together, SVI and CV have expanded the spatial and perceptual scope of walkability research, enabling large-scale, fine-grained assessments of the pedestrian environment.

However, existing methods, whether based on objective measurement or subjective sentiment prediction, remain limited in their ability to guide targeted spatial interventions (Kang et al., 2023). Sentiment-based approaches enable rapid scoring but obscure the spatial mechanisms behind walkability (Blečić et al., 2024). Meanwhile, objective methods grounded in the hierarchy of walking needs offer a theoretical foundation but face two key limitations. First, the precision of feature detection is constrained by current image processing techniques, which often overlook critical elements like power lines or historical landmarks (Koo et al., 2022), let alone the fact that simplistic pixel proportion calculations may fail to capture their perceptual effects (Li et al., 2018). Second, many scoring systems ignore the hierarchical nature of walking decisions, applying linear aggregation methods that may misallocate resources by emphasizing higher-level factors while neglecting unmet basic needs (Li et al., 2022b).

In conclusion, the hierarchy of walking needs serves as the theoretical basis for guiding the improvement of street walkability on the basis of perceptual needs. To effectively enhance the street pedestrian environment, it is essential to fully consider the hierarchical structure of walking to determine the priority of spatial recommendations. Additionally, a more systematic approach is needed to explore the relationships between street environmental factors and spatial perception. Undertaking these two steps of inference is a prerequisite for establishing tailored and actionable spatial improvement recommendations.

Potential of pre-trained large models

The emergence and application of pre-trained large models have shed light on walkability inference, facilitating the transition from diagnosing problems (diagnosis) to determining solutions (treatment)—a core objective in professional practice (Sako, 2024). By integrating domain-specific knowledge, pre-trained large models can offer expert-level solutions and outperform humans in endurance and scalability, making them ideal for large-scale, knowledge-intensive tasks (Yong et al., 2023). Recent attempts have applied pre-trained large models to walkability research, leveraging their multimodal and logical inference capabilities (Zhang et al., 2025). However, these studies remain confined to the feature classification paradigm. Although the use of pre-trained large models improves data annotation efficiency and transferability across regions (Liang et al., 2024), their full potential remains underutilized—particularly their ability to generate multimodal-to-text outputs for professional tasks that extend beyond walkability scoring to improving walking environments.

Our research aims to develop a novel walkability inference model that assesses street walkability and generates tailored and actionable recommendations for pedestrian environment improvement, by fine-tuning and prompting pre-trained large models to integrate domain-specific knowledge of the hierarchy of walking needs.

Model design and implementation

Architecture of the street walkability inference model

In this study, a street walkability inference model is constructed on the basis of the hierarchy of walking needs (Alfonzo, 2005), enabling assessments of the pedestrian environment across different perception levels¹ and providing an overall assessment with spatial recommendations for improving street walkability (Figure 1).

The first step involves filtering street view images to identify urban roads that accommodate pedestrians with visible walking paths. The general knowledge and image recognition capabilities of the pre-trained large model are leveraged, and this step excludes special cases such as nonurban

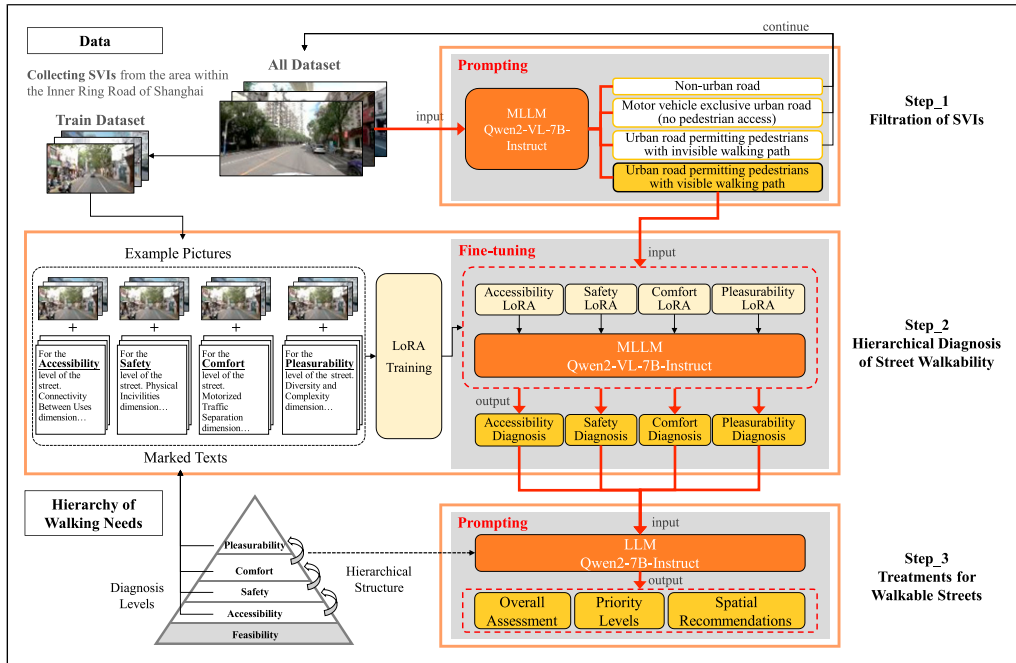


Figure 1. Architecture of the inference model.

roads, motor vehicle-only roads, and sidewalks that are present but invisible. By ensuring that only relevant images are processed, this step enhances the reliability of walkability inference results and prevents nonsensical outputs.

The second step involves a hierarchical diagnosis of street walkability, that is, assessing the extent to which various walking needs are satisfied. By fine-tuning a pre-trained multimodal large language model (MLLM), we encode domain-specific knowledge regarding how environmental factors and their combinations influence individual perceptions. Fine-tuning enhances the general model's ability to avoid overly simplistic or mechanical results, ensuring precise assessments. It enables the model to learn implicit and often vague mappings between environmental factors and subjective perceptions, which are difficult to accurately describe and quantify via natural language (Nasution and Onan, 2024). Additionally, fine-tuning aligns the model with expert perception standards using limited amount of annotated data (Hu et al., 2021). Through low-rank adaptation (LoRA), we train four separate models for each perception level (see Section 3.2 for technical details), enabling independent assessments and spatial issue diagnosis. This provides direct evidence for formulating improvement recommendations.

The third step focuses on inferring treatments for walkable streets. Building on hierarchical diagnosis from the second step, this step aims to identify bottlenecks and propose tailored spatial interventions. Through prompt engineering, hierarchical knowledge on how multiple perception levels jointly influence walking decisions is integrated into the general pre-trained model (see Section 3.3 for technical details), guiding it to generate pedestrian-centric recommendations via a chain-of-thought approach (Wei et al., 2022). The manually designed prompts serve two key functions: clearly transmitting hierarchical rules and providing a structured template for model outputs (Yong et al., 2023). By determining overall assessments and priority levels based on the hierarchy of walking needs, they guide the model to generate spatial recommendations, ensuring

actionable and context-aware interventions. This approach enhances the model's capabilities in semantic learning and logical inference, ensuring professional and coherent outputs.

In the three-step inference process, the pre-trained large models used are from the Qwen2 series². In the first two steps, the Qwen2-VL-7B-Instruct model serves as the base model. It is chosen for its powerful multimodal vision-language inference capabilities. In the third step, the Qwen2-7B-Instruct model is used, leveraging its strengths in text learning and natural language generation. Both models belong to the Qwen2 series, with the Qwen2-VL series providing extended vision-language capabilities and “7B” denoting 7 billion parameters. The “Instruction” model is optimized through supervised fine-tuning (SFT), enhancing its ability to understand and respond to natural language commands. During model inference, we fix the model parameters (Table S1) to ensure stable and consistent results when running the same input across multiple runs.

Fine-tuning a pre-trained MLLM for street walkability diagnosis

To enhance the pre-trained model's performance in domain-specific tasks, the Qwen2-VL-7B-Instruct model is fine-tuned for the hierarchical inference of street walkability. For model specification under limited data conditions, we use low-rank adaptation (LoRA) technology for effective fine-tuning. LoRA enhances the model's performance in text generation tasks related to street walkability by training only low-rank matrices and adding these parameters to the original model. This method avoids updating all the parameters of the base model, thereby reducing the computational and storage overhead, while still modifying the model's behavior without significantly altering its initial performance. The limited parameter changes help reduce training costs, making this method particularly suitable for fine-tuning large language models (Hu et al., 2021).

The hierarchy of walking needs forms the basis for a spatial diagnostic framework. The four levels of walking needs and their corresponding spatial influencing factors are systematically assessed, providing a knowledge base for the four LoRA training sets. The first level, accessibility, concerns connectivity between uses, depending on whether the street depicted in a street view image is passable. The second level, safety, involves two spatial dimensions: physical incivilities, which concern physical disorganization and irregularities in the pedestrian environment, and defensible space features, which focus on design and management measures that enhance pedestrian safety and mitigate potential threats. The third level, comfort, is influenced by three spatial dimensions: motorized traffic separation, referring to designs and measures that buffer pedestrians from vehicle traffic; pedestrian walkway conditions, such as the evenness, width, and quality of sidewalks; and urban design amenities, which include designs and facilities aimed at mitigating the effects of extreme weather and enhancing walking convenience. The fourth level, pleasurability, encompasses four spatial dimensions: diversity and complexity, reflecting variety in architectural facades, building functions, and street elements; liveliness, emphasizing street vibrancy and commercial activities; architectural coherence and scale, focusing on proportionality and harmony; and aesthetic appeal, including building facade quality, cleanliness, and cultural uniqueness.

After identifying the spatial dimensions associated with each perception level, we integrated relevant theories and existing empirical findings to organize the influential environmental factors at each level (Figure 2).³ These factors were classified into positive and negative influences: positive factors enhance the walking experience and promote pedestrian activity, whereas negative factors detract from the quality of the street environment and hinder pedestrian engagement.

To better capture the influence of environmental factors and their interactions on each perception level, we constructed annotated training datasets in the form of image–text pairs. For each perception level, that is, accessibility, safety, comfort, and pleasurability, we selected 200 street view images and invited urban planning experts to annotate them.⁴ Each image was assigned a Likert-scale rating (from “very poor” to “excellent”) and a corresponding natural language justification

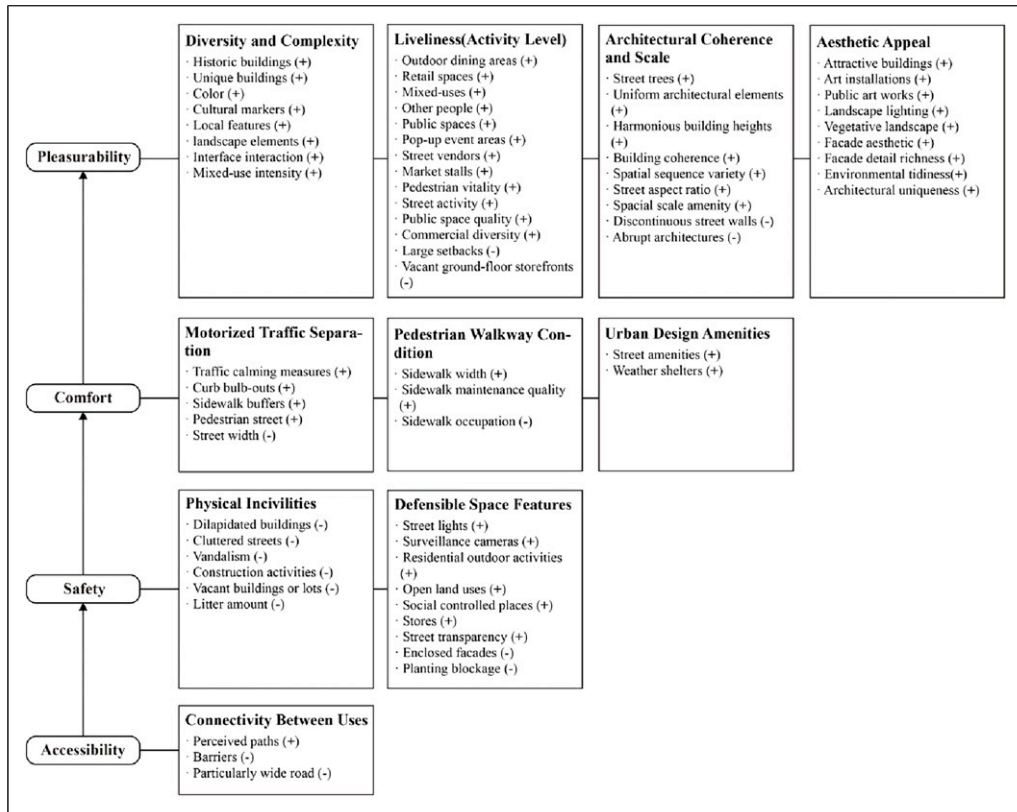


Figure 2. Spatial diagnostic framework based on the hierarchy of walking needs.

grounded in the spatial diagnostic framework. The rating criteria were calibrated through the Delphi method (Brady, 2015) to ensure consistency. The descriptive texts explained how positive and negative environmental factors jointly contributed to the perception rating, thus encoding causal logic that reflects expert reasoning (Table S2). This domain-informed logic not only supports walkability inference with stronger interpretability but also enables the model to generate more context-specific spatial interventions.

These annotated image–text pairs served as training data for fine-tuning the Qwen2-VL-7B-Instruct model using LoRA. Fine-tuning was conducted separately for each perception level, resulting in four specialized LoRA models. The initial learning rate was set to $6.441\text{e-}4$ and gradually decreased throughout the training process. The batch size was 1, with gradient accumulation also set to 1, and the model was trained for 3 epochs. The final model for each level was selected based on the lowest validation loss during training.

Prompting a pre-trained LLM to generate treatments for walkable streets

To realize comprehensive multilevel inference for spatial improvement decision-making, prompt engineering was used to integrate domain-specific knowledge into the Qwen2-7B-Instruct model. The purpose of prompt engineering is to provide a general large model with the necessary instructions and contextual information, such as the theme, settings, roles, and domain-specific

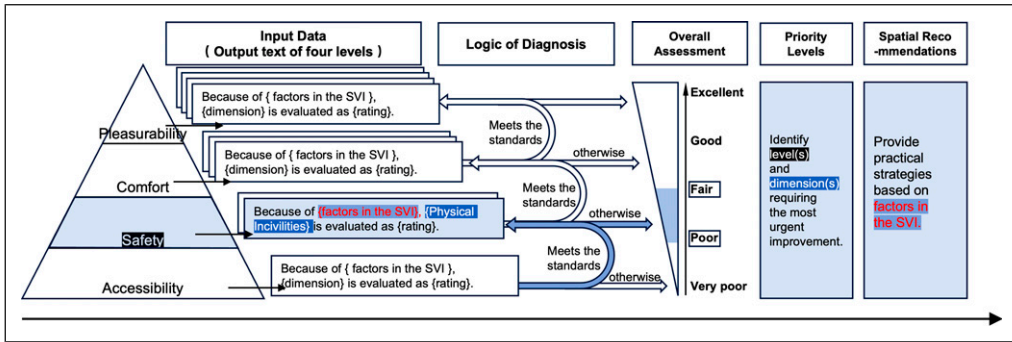


Figure 3. Hierarchical logic in prompt design for street walkability inference.

knowledge (Marvin et al., 2024), thereby enhancing its performance in addressing real-world urban planning challenges.

Figure 3 further illustrates how prompt design embodies the hierarchical structure of walking needs, guiding the overall assessment of street walkability and the derivation of spatial recommendations for pedestrian environments. Specifically, the model processes diagnostic inputs across four levels in a sequential manner. If the basic needs at a lower level are satisfied, the model can proceed to evaluate the higher-level needs. For example, if accessibility needs are met but safety needs are not met, the loop stops, outputting an overall street walkability assessment result and identifying safety as the priority level for improvement. On the basis of this logic, tailored spatial improvement recommendations are then established for the safety level.

To achieve this hierarchical inference logic, we carefully designed the prompts (Table S4), which consist of four components. The input data include the four-level ratings and their corresponding explanations generated in the previous step via the fine-tuned model. The instruction defines the role of an urban planning expert, with the goal of improving the street pedestrian environment. The context provides the theoretical foundation for street walkability inference, grounded in the hierarchy of walking needs theory. The output indicators are divided into three parts: first, generating the overall street walkability rating, ranging from “very poor” to “excellent”; second, identifying the focus of improvement on the basis of hierarchical inference, outputting the level and corresponding dimension to be prioritized; and third, providing specific, actionable spatial strategies for the prioritized level. This prompt design aims to guide the large model in performing chain-of-thought reasoning, with clear guidelines on output requirements and format, ensuring both accuracy and robustness in the results.

Experiments and results

Street walkability assessment

The empirical study area is located within the Inner Ring Road of Shanghai, covering approximately 120 km² and containing 1326 streets (Figure S2). We conducted a rapid, scalable walkability assessment using 17,937 street view images collected from Baidu Street View Map (Figure S3). Ratings across the four levels show a clear hierarchical differentiation (Figure S4). While accessibility and safety are largely ensured, with 91.6% and 82.5% of streets rated as “good” or “better” respectively, there is a marked decline at the comfort (36.1%) and pleasurability (28.8%) levels. Notably, the proportion of streets rated as “poor” or “very poor” increases significantly at the

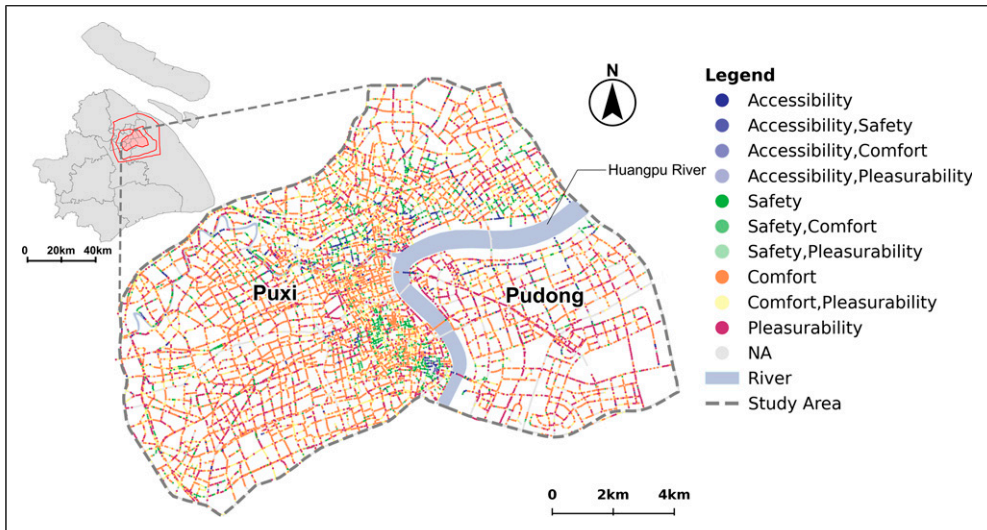


Figure 4. Walkability assessment map for priority improvement level.

comfort level, reaching 25.9%, indicating that discomfort is the most pressing spatial deficiency within Shanghai's Inner Ring Road.

Prioritized improvement needs, derived from hierarchical diagnosis, show clear spatial differentiation (Figure 4). Connectivity and safety deficiencies are concentrated along the west bank of the Huangpu River, where old residential neighborhoods from the 1850s to the 1950s remain. Citywide, comfort and pleasurability are the primary concerns. Puxi, developed earlier, faces issues like limited motorized-nonmotorized traffic separation, poor sidewalks, and a lack of urban amenities, driving widespread demand for comfort improvements. In contrast, Pudong, developed after the 1990s, benefits from spacious sidewalks, clear traffic separation, and abundant greenery, addressing comfort needs. However, its uniform modern business district design reduces pleasurability, lacking diversity, liveliness, and aesthetic appeal. These findings are consistent with our local understanding.

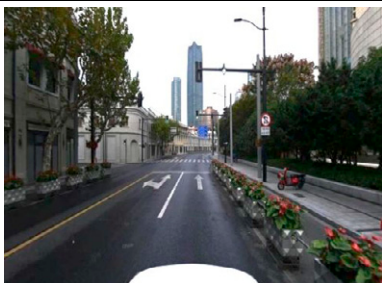
From assessment to improvement

In this section, two typical cases are presented to demonstrate the street walkability inference from hierarchical diagnosis to treatment generation.

Table 1 presents the hierarchical diagnosis results and corresponding treatment for a given street view image.⁵ The model evaluates all four levels, identifying key environmental factors and assessing their impact on walkability, providing a clear and explainable rating. The results show good accessibility and safety, while comfort is rated lower due to insufficient motorized traffic separation. Liveliness is moderate due to a lack of outdoor activity spaces. Based on the hierarchy of walking needs, the model assigns an overall “good” rating, prioritizing comfort and pleasurability for improvement.

Compared with the baseline Qwen2-7B-Instruct model, which was neither fine-tuned nor prompted (Table S5), our model shows marked improvements in clarity, accuracy, and domain relevance. While the baseline model can perform general assessments, it fails to incorporate the hierarchy of walking needs and provides broad, generic suggestions that do not match the actual

Table 1. Walkability inference for Case 1.

Hierarchical diagnostic result	
Accessibility : There are no significant obstacles to pedestrian connectivity. Taking all these factors into account, the overall rating for the accessibility is excellent .	
Safety: The urban physical environment shows no significant disorder or irregularities, leading to a rating of excellent for physical incivilities . The presence of open spaces such as squares and pocket parks, along with shops, high ground-level transparency, and a secure street environment, results in a rating of excellent for defensible space features . Taking all these factors into account, the overall rating for safety is excellent .	
Comfort: The street is relatively narrow, and the raised curbstone separates the sidewalk from motor vehicles. However, the absence of traffic-calming measures leads to the motorized traffic separation dimension being rated as fair . The sidewalk is wide, free from obstructions, and well-maintained, resulting in the pedestrian walkway condition dimension being rated as excellent . Neatly arranged street trees create continuous shade, providing a shaded walking environment. Additionally, landscaping features along the sidewalk improve the street's micro-environment, leading to the urban design amenities dimension being rated as excellent . Taking all these factors into account, the overall rating for comfort is good .	
Pleasurability: The combination of historical and modern buildings, good visual permeability, and strong interface interaction results in a rating of excellent for diversity and complexity . The lack of prominent outdoor activity spaces or noticeable human activity leads to a rating of fair for activity level . The consistent architectural style and the presence of street trees contributing to coordination result in a rating of good for architectural coherence and scale . The clean street environment, green decorations, and attractive architectural style result in a rating of excellent for aesthetic appeal . Taking all these factors into account, the overall rating for pleasurability is good .	
Treatment report	
Overall Assessment: The overall rating of the pedestrian environment is good .	
Priority Levels: Highest level for improvement: Considering that the accessibility and safety scores have reached a high level, and the comfort and pleasurability scores are good, there is still some room for improvement, we should focus on comfort and pleasurability improvements.	
Spatial Recommendations: (See Table S5 for details.)	

street context, such as unnecessary greenbelt improvements. In contrast, our walkability inference model produces more reliable and specialized outputs. For instance, in response to insufficient street liveliness, it recommends leveraging local historical architecture to develop cultural pathways or tourist routes, demonstrating expert-level awareness of Shanghai’s urban fabric.

Table 2 involves a street view image (Case 2) from a traditional residential area in Shanghai, assessed as “fair” overall. The model concludes that, despite shortcomings in pleasurability, safety and comfort must be addressed first due to the hierarchy of walking needs. The comparison with the baseline Qwen2-7b-Instruct model (without fine-tuning and prompting) and its prompted version (with prompting only) is further explored (Table S6), demonstrating the professional depth and high adaptability of our street walkability inference model.

These results validate the effectiveness of fine-tuning and prompt engineering in integrating domain-specific knowledge into a general-purpose large model, enabling accurate spatial diagnostics and tailored recommendations across 17,937 street view images in Shanghai’s Inner Ring Road. This directly informs the formulation of walkability-oriented street renewal policies by assisting in the rapid identification of streets most in need of renewal, providing zonal spatial improvement strategies at the macro level, and offering urban designers rich spatial insights and practical ideas for implementing the “one-site, one-solution” renewal project.

Model validation

To systematically evaluate the performance of the proposed street walkability inference model, 13 volunteers with an educational background in urban and rural planning (master’s degree or

Table 2. Walkability inference for Case 2.

Hierarchical diagnostic result	
Accessibility : There are no significant obstacles to pedestrian connectivity. Taking all these factors into account, the overall rating for the accessibility is excellent .	
Safety : The presence of overhead wires and a cluttered street environment, along with minor spatial disorder and irregularities, results in the physical incivilities dimension being rated as fair . The presence of shops and a partially enclosed interface enhances safety and provides protection against potential threats, leading to a defensible space features dimension rating of good . Taking all these factors into account, the overall rating for Safety is fair .	
Comfort : The narrow street and low traffic volume provide pedestrians with a sense of safety while walking. The raised curbstone effectively separates the sidewalk from motor vehicles, reducing pedestrian-vehicle conflicts. However, the lack of traffic-calming measures to slow down motor vehicles results in the motorized traffic separation dimension being rated as poor . Sidewalks are narrow and cluttered with disorganized non-motorized vehicles, leading to a pedestrian walkway condition dimension rating of poor . Additionally, the absence of street trees, shaded areas, sun shelters, resting facilities, and landscaping features results in the urban design amenities dimension being rated as very poor . Taking all these factors into account, the overall rating for Comfort is poor .	
Pleasurability : The diverse commercial shops and architectural styles, combined with utility poles and wires, add visual complexity, leading to a diversity and complexity dimension rating of good . The presence of pedestrian and vehicular activity provides a sense of vibrancy, resulting in activity level dimension rating of good . However, while the architectural style is relatively uniform, utility poles and wires disrupt consistency, so the architectural coherence and scale dimension is rated as fair . Furthermore, utility poles and wires contribute to visual clutter, and building facades lack distinctiveness, resulting in the aesthetic appeal dimension being rated as fair . Taking all these factors into account, the overall rating for Pleasurability is fair .	Treatment report Overall Assessment: The overall rating of the walking environment is Fair . Priority Levels: The most critical areas for improvement are Comfort and Safety . Among these, comfort has the lowest rating, primarily due to the lack of traffic calming measures and insufficient pedestrian sidewalk width. Safety follows closely, with issues such as spatial disorder caused by overhead wires.
	Spatial Recommendations: (See Table S6 for details.)

higher) were invited to validate the model's inference results. A test set of 100 street view images was constructed through stratified random sampling. The validation aimed to reveal the extent to which the model achieves consensus with human experts. It's not only reflected in the consistency of hierarchical spatial perception but, more importantly, in the agreement on priority improvement levels. On this basis, the usefulness, specificity, and feasibility of the model's proposed spatial improvement recommendations were evaluated using a five-point Likert scale. The goal was to reveal the extent to which the model can serve as a potential substitute for human experts in supporting large-scale spatial improvement practices.

Table 3 indicates no systematic bias between the model's hierarchical assessment compared to human experts. Spearman rank correlation analysis shows a high level of ordinal consistency between the model and expert ratings. Standard deviation analysis reveals that over 60% of the model's outputs fall within the experts' ± 1 standard deviation range at each perception level. Fixed-effects analysis from the linear mixed-effects model (LMM) further confirms that, after controlling for random effects from images and experts, there is no significant difference between the model's scores and those of the experts.

Regarding priority improvement level judgment, the model performs comparably to human experts (Table 4). Figure S5 illustrates natural variability among experts, reflecting subtle perceptual differences. Despite this variability, the model consistently aligns with the average expert's performance across subset accuracy and Hamming loss metrics. This indicates the model's strong potential to effectively support spatial improvement decision-making.

Table 3. Hierarchical assessment validation.

		Accessibility	Safety	Comfort	Pleasurability
Spearman rank correlation analysis	Coefficient (ρ)	0.494	0.753	0.768	0.776
	p -value	1.79e-07***	1.6e-19***	1.23e-20***	2.51e-21***
Standard deviation analysis	Proportion within ± 1 SD	85.0%	74.0%	63.0%	67.0%
Fixed-effects analysis (LMM)	Coefficient (β)	0.01385	-0.06154	-0.3023	-0.06154
	p -value	0.876	0.768	0.300	0.780

The evaluation results show that the spatial improvement recommendations generated by the model have been generally recognized by experts, with the average scores for the three dimensions all above the neutral level (Figure S6). Specificity received the highest rating (mean = 4.21, SD = 0.50), followed by usefulness (mean = 3.88, SD = 0.60), while feasibility was rated relatively lower (mean = 3.57, SD = 0.74). This may be due to the fact that the spatial diagnosis in the second step focused only on the environmental factors defined by the established framework (Figure 2), which limited the spatial context available for treatment generation and excluded broader information such as road functions and traffic conditions. Additionally, socio-economic costs related to spatial policy implementation were not fully considered. Nevertheless, the street walkability inference model developed in this study successfully identifies key spatial issues in specific street view images and provides tailored recommendations for improving walkability. While some suggestions may require further refinement in terms of feasibility, the proposed model can offer valuable insights in large-scale real-world applications.

Discussion

This study employs fine-tuning and prompt engineering for pre-trained large models to diagnose street walkability and generate improvement recommendations for walkable streets. Building upon the widespread use of street view imagery for large-scale street walkability scoring and comparison, this approach offers two key advantages to better inform spatial governance. First, it provides clear explanations for the ratings, facilitating tailored and professional improvement recommendations. By integrating domain-specific knowledge, the model captures how environmental factors influence subjective perceptions and considers hierarchical decision-making rules, generating prioritized interventions that align with pedestrians' needs. Second, it leverages the multimodal understanding and logical reasoning capabilities of large models, significantly reducing training costs. Unlike traditional methods requiring thousands of labeled images, this model achieves domain adaptation with as few as two hundred images, matching human expert performance.

The core contribution of this study lies in proposing a technical framework for using pre-trained multimodal large models for street walkability inference, achieving a breakthrough from large-scale scoring to interpretable spatial problem diagnosis and tailored treatment generation. This approach

Table 4. Priority improvement level judgment validation.

	Model	Expert_avg
Subset accuracy	44.00%	43.38%
Hamming loss	20.00%	18.83%

could also be extended to broader urban design theories such as urban imagery and urban vitality, leveraging the integration of large model technology with vast street view imagery to support large-scale urban planning practices, from urban spatial perception to diagnosis and improvement.

Despite these advances, the study has several limitations. First, the fine-tuning training set used in this study is based on Shanghai, providing a deep understanding of local streetscapes and supporting localized spatial practices. However, this focus may limit the model's generalizability to other regions and cities (Arellana et al., 2020). Second, the study did not fully account for demographic heterogeneity. Future research exploring the walking needs and preferences of different demographic groups could promote more inclusive and adaptive development (Lieu and Guhathakurta, 2025; Mazzulla et al., 2024).

While our model could not directly be adapted across varying socio-cultural and morphological contexts beyond Shanghai, the street walkability inference framework proposed, comprising street view image filtration, hierarchical walkability diagnosis, and treatment generation, remains highly transferable. By conducting local empirical research and demographic demand surveys to map the “pedestrian environment–spatial perception–walking decision” knowledge, local label sets can be constructed and pre-trained large models can be fine-tuned to support street walkability diagnosis and optimization tailored to local needs. In this way, the framework can guide cost-effective, high-performance fine-tuning and prompt engineering of pre-trained large models, supporting localized street walkability improvement designs.

This study represents an initial exploration into using pre-trained large models to support street pedestrian environment improvement, with substantial potential for future advancement. Currently, the model generates spatial recommendations in natural language, but it is not yet able to precisely evaluate to which extent these treatments could improve street walkability or increase pedestrians' willingness to walk. Future research could leverage the text-to-image generation capabilities of large models to visualize the spatial outcomes of proposed interventions (Huang and Oki, 2025). Such visualizations would enable more intuitive assessments of their effectiveness and feasibility, support iterative refinement of spatial design schemes, and promote public participation.

Conclusion

This study proposes a street walkability inference framework that leverages pre-trained large models to diagnose and improve pedestrian environments. Fine-tuning and prompt engineering provide a technical pathway for integrating domain-specific knowledge, enabling expert-level performance and professional outputs. As a result, the developed street walkability inference model not only enables rapid, large-scale, and high-precision assessments but also clearly explains why a street performs poorly in relation to the hierarchy of walking needs theory. This allows for precise spatial diagnosis and tailored treatment generation from the pedestrian's perspective. As a promising pathway for advancing walkability research, this approach warrants further exploration in both application contexts and technical development.

ORCID iDs

Shijia Lin  <https://orcid.org/0009-0005-0087-6162>

Xinyi Niu  <https://orcid.org/0000-0001-6611-0331>

Siwei Zhou  <https://orcid.org/0009-0008-7390-0671>

Yeyoumin Tian  <https://orcid.org/0000-0002-3626-2331>

Chenyi Wang  <https://orcid.org/0009-0000-4038-6700>

Mingqi Shu  <https://orcid.org/0009-0004-6841-4100>

Author contributions

Shijia Lin: Conceptualization, methodology, validation, project administration, writing—original draft, and writing—review and editing. Xinyi Niu: Conceptualization, methodology, funding acquisition, resources, and supervision. Siwei Zhou: Software, formal analysis, visualization, and writing—original draft. Yeyoumin Tian: Data curation, formal analysis, visualization, and writing—original draft. Chenyi Wang: Formal analysis, visualization, and writing—original draft. Mingqi Shu: Formal analysis, visualization, and writing—original draft. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supplemental Material

Supplemental material for this article is available online.

Notes

1. Given that this study aims to achieve street walkability inference based on street view imagery, it focuses solely on the perception levels influenced by the street environment in the hierarchy of walking needs, namely, accessibility, safety, comfort, and pleurability ([Figure S1 in the Supplemental Material](#)).
2. The Qwen2 series model is an open-source model that was first provided by the Alibaba Cloud Tongyi Qianwen team in 2024, offering strong learning and expression abilities for both Chinese and English text. It has outperformed Meta's Llama-3-70B in several benchmark evaluations. In this study, Chinese is selected as the language for both training and testing, and the text is translated into English.
3. [Table S3](#) provides detailed references for all environmental factors associated with each level and dimension.
4. The expert annotation process is described in detail in the Supplementary Materials.
5. The hierarchical diagnostic result is the output from the fine-tuned multimodal large model in the second step, which serves as input for the third step. The treatment report, in contrast, is the output from the prompted large language model in the third step.

References

- Alfonzo MA (2005) To walk or not to walk? The hierarchy of walking needs. *Environment and Behavior* 37(6): 808–836.
- Alidoust S, Bosman C and Holden G (2018) Talking while walking: an investigation of perceived neighbourhood walkability and its implications for the social life of older people. *Journal of Housing and the Built Environment* 33(1): 133–150.

- Arellana J, Saltařın M, Larrařaaga AM, et al. (2020) Urban walkability considering pedestrians' perceptions of the built environment: a 10-year review and a case study in a medium-sized city in Latin America. *Transport Reviews* 40(2): 183–203.
- Blećić I, Saiu Vand A and Trunfio G (2024) Enhancing urban walkability assessment with multimodal large language models. In: Gervasi O, Murgante B and Garau C (eds) *Computational Science and Its Applications – ICCSA 2024 Workshops*. Springer Nature Switzerland, pp. 394–411.
- Brady SR (2015) Utilizing and adapting the Delphi method for use in qualitative research. *International Journal of Qualitative Methods* 14(5): 1609406915621381.
- Cohen A, Dalyot S, Natapov A, et al. (2024) How accessible are cities for visually impaired pedestrians? A case of greater London. *Environment and Planning B: Urban Analytics and City Science* 51(7): 1614–1631.
- De Vos J, Lättman K, van der Vlught A-L, et al. (2023) Determinants and effects of perceived walkability: a literature review, conceptual model and research agenda. *Transport Reviews* 43(2): 303–324.
- Ewing R and Handy S (2009) Measuring the unmeasurable: urban design qualities related to walkability. *Journal of Urban Design* 14(1): 65–84.
- Fonseca F, Ribeiro PJG, Conticelli E, et al. (2021) Built environment attributes and their influence on walkability. *International Journal of Sustainable Transportation* 16(7): 660–679.
- Gan Z, Yang M, Zeng Q, et al. (2021) Associations between built environment, perceived walkability/bikeability and metro transfer patterns. *Transportation Research Part A: Policy and Practice* 153: 171–187.
- Hajrasouliha A and Yin L (2015) The impact of street network connectivity on pedestrian volume. *Urban Studies* 52(13): 2483–2497.
- Handy SL, Boarnet MG, Ewing R, et al. (2002) How the built environment affects physical activity: views from urban planning. *American Journal of Preventive Medicine* 23(2 Supplement 1): 64–73.
- He X and He SY (2023) Using open data and deep learning to explore walkability in shenzhen, China. *Transportation Research Part D: Transport and Environment* 118: 103696.
- He X and He SY (2024) Using mobile phone big data and street view images to explore the mismatch between walkability and walking behavior. *Transportation Research Part A: Policy and Practice* 180: 103946.
- Hu EJ, Shen Y, Wallis P, et al. (2021) LoRA: low-rank adaptation of large language models. arXiv: 2106.09685.
- Huang L and Oki T (2025) Enhancing people's walking preferences in street design through generative artificial intelligence and crowdsourcing surveys: the case of Tokyo. *Environment and Planning B: Urban Analytics and City Science* 52: 23998083251327405.
- Kang Y, Kim J, Park J, et al. (2023) Assessment of perceived and physical walkability using street view images and deep learning technology. *ISPRS International Journal of Geo-Information* 12(5): 186.
- Koo BW, Guhathakurta S and Botchwey N (2022) How are neighborhood and street-level walkability factors associated with walking behaviors? A big data approach using street view images. *Environment and Behavior* 54(1): 211–241.
- Li Y, Peng L, Wu C, et al. (2022a) Street view imagery (SVI) in the built environment: a theoretical and systematic review. *Buildings* 12(8): 1167.
- Li X, Santi P, Courtney TK, et al. (2018) Investigating the association between streetscapes and human walking activities using google street view and human trajectory data. *Transactions in GIS* 22(4): 1029–1044.
- Li Y, Yabuki N and Fukuda T (2022b) Measuring visual walkability perception using panoramic street view images, virtual reality, and deep learning. *Sustainable Cities and Society* 86: 104140.
- Liang H, Zhang J, Li Y, et al. (2024) Automatic estimation for visual quality changes of street space via street-view images and multimodal large language models. *IEEE Access* 12: 87713–87727.
- Lieu SJ and Guhathakurta S (2025) Exploring pedestrian route choice preferences by demographic groups: analysis of street attributes in Chicago. *Transportation Research Part A: Policy and Practice* 195: 104437.
- Marvin G, Hellen N, Jjingo D, et al. (2024) Prompt engineering in large language models. In: Jacob IJ, Pıramuthu S and Falkowski-Gilski P (eds) *Data Intelligence and Cognitive Informatics*. Springer Nature, pp. 387–402.

- Mazzulla G, Eboli L and Forciniti C (2024) Do women perceive pedestrian path attractiveness differently from men? *Transportation Research Part A: Policy and Practice* 179: 103890.
- Naik N, Philipoom J, Raskar R, et al. (2014) Streetscore – predicting the perceived safety of one million streetscapes. In: 2014 IEEE Conference on Computer Vision and Pattern Recognition Workshops, Columbus, Ohio, 23 June–28 June 2014, pp. 793–799.
- Nasution AH and Onan A (2024) ChatGPT label: comparing the quality of human-generated and LLM-generated annotations in low-resource language NLP tasks. *IEEE Access* 12: 71876–71900.
- Park S, Choi K and Lee JS (2015) To walk or not to walk: testing the effect of path walkability on transit users' access mode choices to the station. *International Journal of Sustainable Transportation* 9(8): 529–541.
- Root ED, Silbernagel K and Litt JS (2017) Unpacking healthy landscapes: empirical assessment of neighborhood aesthetic ratings in an urban setting. *Landscape and Urban Planning* 168: 38–47.
- Sako M (2024) How generative AI fits into knowledge work. *Communications of the ACM* 67(4): 20–22.
- Song Q, Fang Y, Li M, et al. (2025) Exploring the coherence and divergence between the objective and subjective measurement of streetscape perceptions at the neighborhood level: a case study in shanghai. *Environment and Planning B: Urban Analytics and City Science* 52(5): 1231–1251.
- Trucco E, Verri A and Internet A (1998) *Introductory Techniques for 3-D Computer Vision*. Prentice Hall.
- Wang H and Yang Y (2019) Neighbourhood walkability: a review and bibliometric analysis. *Cities* 93: 43–61.
- Wang M, Haworth J, Chen H, et al. (2024) Investigating the potential of crowdsourced street-level imagery in understanding the spatiotemporal dynamics of cities: a case study of walkability in inner London. *Cities* 153: 105243.
- Wei J, Wang X, Schuurmans D, et al. (2022) Chain-of-Thought prompting elicits reasoning in large language models. arXiv: 2201.11903.
- Yang J, Fricker P and Jung A (2024) From intangible to tangible: the role of big data and machine learning in walkability studies. *Computers, Environment and Urban Systems* 109: 102087.
- Ye Y, Richards D, Lu Y, et al. (2019) Measuring daily accessed street greenery: a human-scale approach for informing better urban planning practices. *Landscape and Urban Planning* 191: 103434.
- Yin L and Wang Z (2016) Measuring visual enclosure for street walkability: using machine learning algorithms and google street view imagery. *Applied Geography* 76: 147–153.
- Yong G, Jeon K, Gil D, et al. (2023) Prompt engineering for zero-shot and few-shot defect detection and classification using a visual-language pretrained model. *Computer-Aided Civil and Infrastructure Engineering* 38(11): 1536–1554.
- Zhang J, Li Y, Fukuda T, et al. (2025) Urban safety perception assessments via integrating multimodal large language models with street view images. *Cities* 165: 106122.
- Zhang F, Zhou B, Liu L, et al. (2018) Measuring human perceptions of a large-scale urban region using machine learning. *Landscape and Urban Planning* 180: 148–160.

Author biographies

Shijia Lin is a Ph.D. Candidate at the College of Architecture and Urban Planning, Tongji University. Her research interests include urban spatial structure and spatiotemporal big-data analytics, along with generative-AI applications for planning diagnostics, inference, and decision support.

Xinyi Niu is a Professor and Doctoral Supervisor at the College of Architecture and Urban Planning, Tongji University, and Deputy Director of the Key Laboratory of Spatial Intelligent Planning Technology, Ministry of Natural Resources of the People's Republic of China. His research expertise centers on urban planning technologies and methods, as well as theories and methodologies of regional spatial development.

Siwei Zhou is a Master's student at the College of Architecture and Urban Planning, Tongji University. Her research interests focus on AI applications in urban planning, particularly urban renewal.

Yeyoumin Tian is a Master's student at the College of Architecture and Urban Planning, Tongji University. His research interests revolve around urban planning technologies and methods, with a focus on integrating artificial intelligence with traditional planning and exploring innovative applications of large models in spatial planning and design.

Chenyi Wang is a Master's student at the College of Architecture and Urban Planning, Tongji University. His research interests lie in urban planning technology and methods, with a specific focus on AI-assisted urban spatial data analysis and its applications.

Mingqi Shu is a Master's student at the College of Architecture and Urban Planning, Tongji University. Her research interests include regional-scale urban studies and community-scale urban renewal design.